EXPLORATORY DATA ANALYSIS (EDA) ON TITANIC DATASET

DATA ANALYST INTERNSHIP – TASK 5

Submitted By

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Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Dataset

Titanic Dataset

ABSTRACT

Exploratory Data Analysis (EDA) is a fundamental step in the data analysis process that helps analysts understand the structure, quality, and characteristics of a dataset. The purpose of this project is to perform a detailed Exploratory Data Analysis on the Titanic Dataset, one of the most widely used datasets in data science and machine learning.

The analysis focuses on identifying patterns, trends, correlations, and anomalies within passenger information. Various statistical techniques and visualization methods were employed using Python libraries such as Pandas, Matplotlib, and Seaborn. Through the analysis, important factors influencing passenger survival were identified, including gender, passenger class, age, and fare amount.

The insights obtained from this project demonstrate how EDA can transform raw data into meaningful information and support data-driven decision-making.

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1. INTRODUCTION

The Titanic disaster remains one of the most well-known maritime tragedies in history. On April 15, 1912, the RMS Titanic sank after colliding with an iceberg during its maiden voyage. The disaster resulted in the loss of more than 1,500 lives.

The Titanic dataset contains passenger information and survival records, making it an ideal dataset for learning data analysis techniques. The objective of this project is to analyze the dataset and discover the factors that contributed to passenger survival.

Exploratory Data Analysis helps in:

- Understanding dataset characteristics
- Identifying missing values
- Detecting outliers
- Discovering relationships among variables
- Supporting predictive modeling

This report presents a comprehensive analysis of the Titanic dataset using statistical summaries and visualizations.

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2. OBJECTIVES OF THE PROJECT

The main objectives of this project are:

1. To understand the structure of the Titanic dataset.
2. To identify missing values and data quality issues.
3. To perform data cleaning and preprocessing.
4. To explore statistical properties of the data.
5. To visualize patterns and trends.
6. To identify factors affecting passenger survival.
7. To generate meaningful insights through EDA.

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3. DATASET DESCRIPTION

The Titanic dataset contains information about passengers aboard the RMS Titanic.

Dataset Size

- Number of Rows: 891
- Number of Columns: 12

Features Description

Feature	Description
PassengerId	Unique Passenger ID
Survived	Survival Status
Pclass	Passenger Class
Name	Passenger Name

Feature	Description
Sex	Gender
Age	Age of Passenger
SibSp	Siblings/Spouses Aboard
Parch	Parents/Children Aboard
Ticket	Ticket Number
Fare	Ticket Fare
Cabin	Cabin Number
Embarked	Port of Embarkation

Target Variable

Survived:

- 0 = Did Not Survive
- 1 = Survived

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4. DATA PREPROCESSING

Before performing analysis, the dataset was examined for missing values and inconsistencies.

Missing Values Analysis

The following columns contained missing values:

- Age
- Cabin
- Embarked

Data Cleaning Steps

1. Missing values in Age were replaced with the median age.
2. Missing values in Embarked were replaced with the most frequent category.
3. Cabin column was removed because a large percentage of values were missing.

Importance of Data Cleaning

Data cleaning improves dataset quality and ensures that visualizations and statistical results are accurate and reliable.

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5. EXPLORATORY DATA ANALYSIS

5.1 Statistical Summary

The dataset was analyzed using:

- df.info()
- df.describe()
- df.value_counts()

Findings

- Average passenger age is approximately 29 years.
- Most passengers traveled in Third Class.
- Fare values vary significantly.
- Several numerical variables contain extreme values.

5.2 Age Distribution Analysis

A histogram was plotted to study the age distribution.

Observations

- Majority of passengers were between 20 and 40 years old.
- Few passengers were older than 70 years.
- Distribution is slightly skewed.

Interpretation

The dataset mainly represents young and middle-aged passengers.

5.3 Fare Distribution Analysis

A boxplot was used to examine ticket fare values.

Observations

- Significant outliers are present.
- Some passengers paid exceptionally high fares.
- Fare distribution is highly right-skewed.

Interpretation

Passengers paying higher fares generally belonged to higher classes.

5.4 Survival Distribution

A count plot was generated for survival status.

Observations

- More passengers died than survived.
- Survival percentage is lower than mortality percentage.

Interpretation

The disaster had a severe impact on passenger survival.

5.5 Survival by Gender

A count plot comparing gender and survival was created.

Observations

- Female passengers survived at a much higher rate.
- Male passengers experienced higher mortality.

Interpretation

Women were given evacuation priority, leading to higher survival rates.

5.6 Survival by Passenger Class

Passenger class was analyzed against survival status.

Observations

- First-Class passengers had the highest survival rate.
- Third-Class passengers had the lowest survival rate.

Interpretation

Access to lifeboats and location within the ship likely influenced survival chances.

5.7 Age vs Fare Analysis

A scatter plot was used to study the relationship between Age and Fare.

Observations

- No strong linear relationship exists.
- Higher fare values are concentrated among First-Class passengers.

Interpretation

Fare is more related to passenger class than passenger age.

5.8 Correlation Analysis

A heatmap was generated to examine correlations among numerical variables.

Observations

- Fare positively correlates with survival.
- Passenger class negatively correlates with survival.
- Age shows weak correlation with survival.

Interpretation

Passenger Class and Fare are stronger predictors of survival than Age.

5.9 Pairplot Analysis

A pairplot was generated to visualize relationships among multiple variables.

Observations

- Clear survival patterns emerge for Fare and Passenger Class.
- Age distributions overlap considerably.

Interpretation

Passenger Class and Fare provide better discrimination between survivors and non-survivors.

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6. KEY INSIGHTS

The following insights were derived from the analysis:

Insight 1

Gender significantly influenced survival outcomes.

Insight 2

First-Class passengers had higher chances of survival compared to Second and Third-Class passengers.

Insight 3

Most passengers belonged to Third Class.

Insight 4

Fare is positively associated with survival probability.

Insight 5

Age alone is not a strong predictor of survival.

Insight 6

The dataset contains several fare outliers.

Insight 7

Data preprocessing was necessary due to missing values.

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7. CHALLENGES FACED

During the project, several challenges were encountered:

- Missing values in Age and Cabin columns.
- Highly skewed Fare distribution.
- Presence of outliers.
- Large number of missing Cabin entries.

These challenges were addressed through data cleaning and preprocessing techniques.

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8. CONCLUSION

The Exploratory Data Analysis conducted on the Titanic dataset successfully revealed important patterns and relationships within the data.

The analysis showed that Gender, Passenger Class, and Fare were the most influential factors affecting passenger survival. Female passengers and First-Class passengers had significantly better survival chances compared to male passengers and Third-Class passengers.

Although Age contributed some information, it was not as influential as the other variables. The project demonstrated the effectiveness of statistical analysis and visualization techniques in extracting meaningful insights from real-world datasets.

The results obtained from this study provide a strong foundation for future predictive modeling and machine learning applications.

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9. LEARNING OUTCOMES

Through this project, the following skills were developed:

- Data Collection and Understanding
- Data Cleaning and Preprocessing
- Exploratory Data Analysis
- Statistical Interpretation
- Data Visualization
- Correlation Analysis
- Insight Generation
- Python Programming using Pandas and Seaborn

This project successfully achieved its objective of identifying patterns, trends, relationships, and anomalies in the Titanic dataset through comprehensive exploratory analysis.